Screenshots — Terraform Data Sources

Lecture 7 · 9 shots, chronological · source stills from the session (2026-10-05 22:25 – 23:07)

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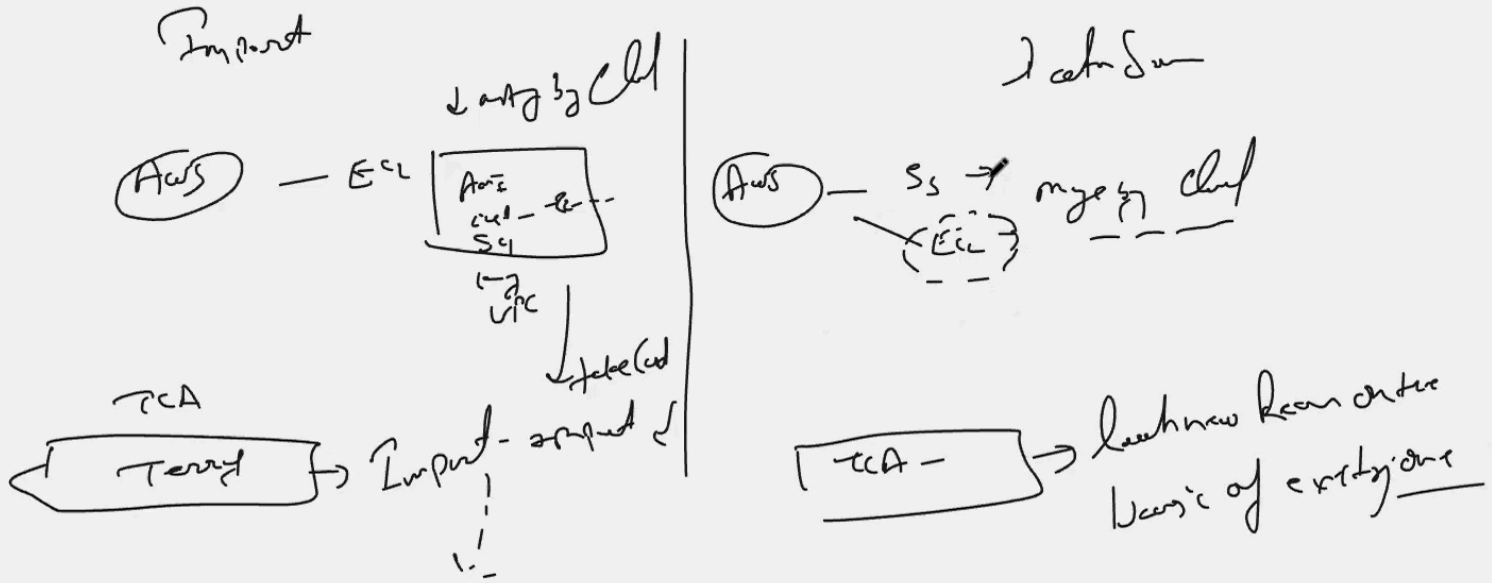


In Terraform, a **data source** allows a configuration to fetch or compute information defined outside of Terraform, or from another distinct Terraform workspace. It serves as a **read-only view** of your infrastructure, enabling you to reference existing resources without modifying or managing them. GeeksforGeeks +2

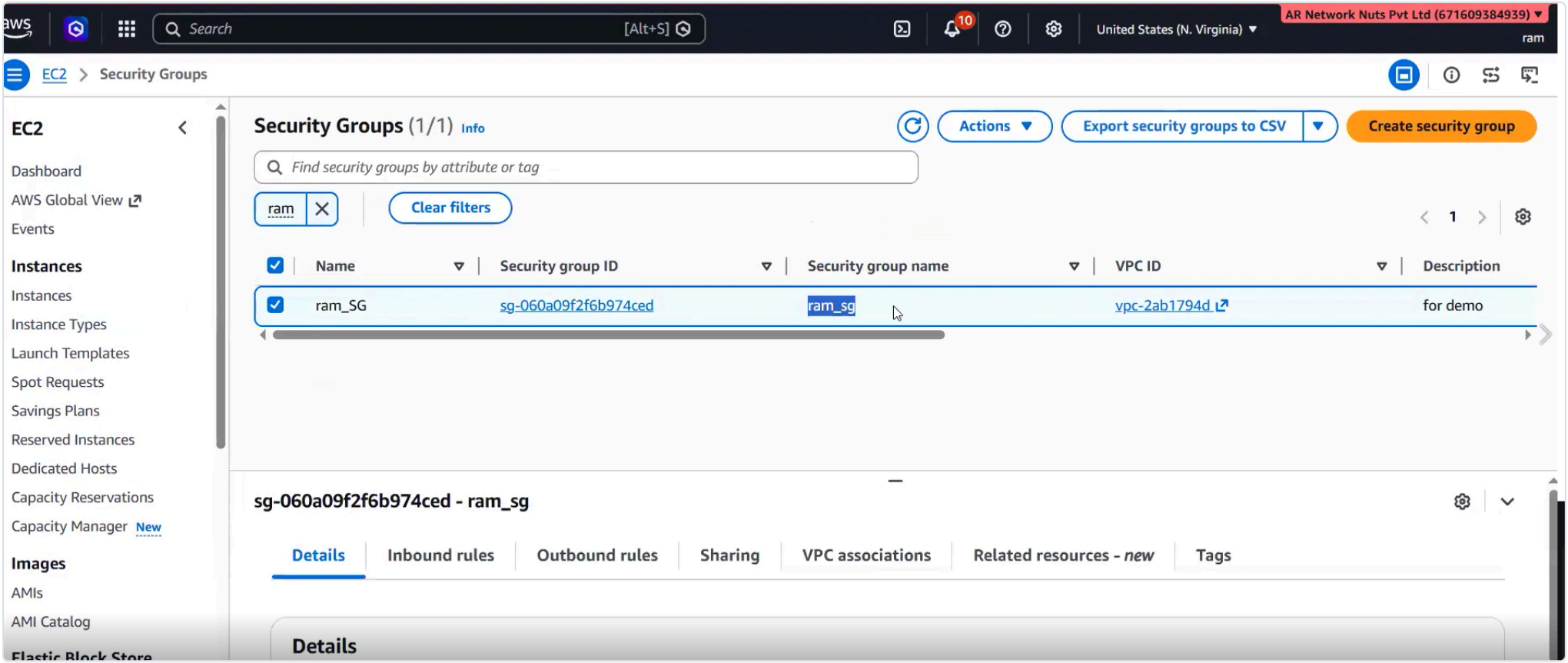
miro Untitled Upgrade Share

Data Sources

In Terraform, a **data source** allows a configuration to fetch or compute information defined outside of Terraform, or from another distinct Terraform workspace. It serves as a **read-only view** of your infrastructure, enabling you to reference existing resources without modifying or managing them



AWS console — existing security group `ram_sg` = `sg-060a09f2f6b974ced` in `vpc-2ab1794d` (us-east-1)




root@Devops-server:~/project1 — vim main.tf
provider "aws" {
region = "us-east-1"
}
Look up an existing security group by name in NV region
data "aws_security_group" "example" {
name = "ram_sg"
}
output "my-ouput" {
value = data.aws_security_group.example.id
}

```
[root@Devops-server project1]# terraform apply --auto-approve
data.aws_security_group.example: Reading...
data.aws_security_group.example: Read complete after 2s [id=sg-060a09f2f6b974ced]

Changes to Outputs:
  + my-ouput = "sg-060a09f2f6b974ced"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

my-ouput = "sg-060a09f2f6b974ced"
[root@Devops-server project1]#
```

```
# Look up an existing security group by name in NV region
data "aws_security_group" "example" {
  name = "ram_sg"
}

output "my-ouput" {
value = data.aws_security_group.example.id
}

# Launch an EC2 instance and associate it with the existing security group in NV region
resource "aws_instance" "example" {
  ami           = "ami-05ffe3c48a9991133" # Replace with a real AMI ID
  instance_type = "t2.micro"
  key_name      = "ram_nv"
  vpc_security_group_ids = [data.aws_security_group.example.id]
  tags = {
    Name = "NV_server"
    source = "terraform"
  }
}
```

```
data "aws_ami" "ubuntu" {
  most_recent = true

  filter {
    name   = "name"
    values = ["ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-*"]
  }

  filter {
    name   = "virtualization-type"
    values = ["hvm"]
  }

  owners = ["099720109477"] # Canonical
}

resource "aws_instance" "example" {
  ami           = data.aws_ami.ubuntu.id
  instance_type = "t3.micro"

  tags = {
    Name = "HelloWorld"
  }
}
```

root@Devops-server:~/project1 — vim data.tf

```
# Define a data source to fetch the latest Amazon Linux 2 AMI
data "aws_ami" "latest_amazon_linux" {
  most_recent = true
  owners = ["amazon"]
  filter {
    name   = "name"
    values = ["amzn2-ami-hvm-*-x86_64-gp2"]
  }
}
output "my-ouput" {
value = data.aws_ami.latest_amazon_linux.id
}

~
~
~
~
```